$$P_M = MM^+, \quad \widehat{Y} = YP_M, \quad E = Y - \widehat{Y},$$
$$D_{\text{space}} = \frac{\|E\|_F^2}{\|Y\|_F^2}, \qquad D_{\text{rank},K} = \frac{\sum_{\ell>K} \sigma_\ell^2}{\sum_\ell \sigma_\ell^2}.$$